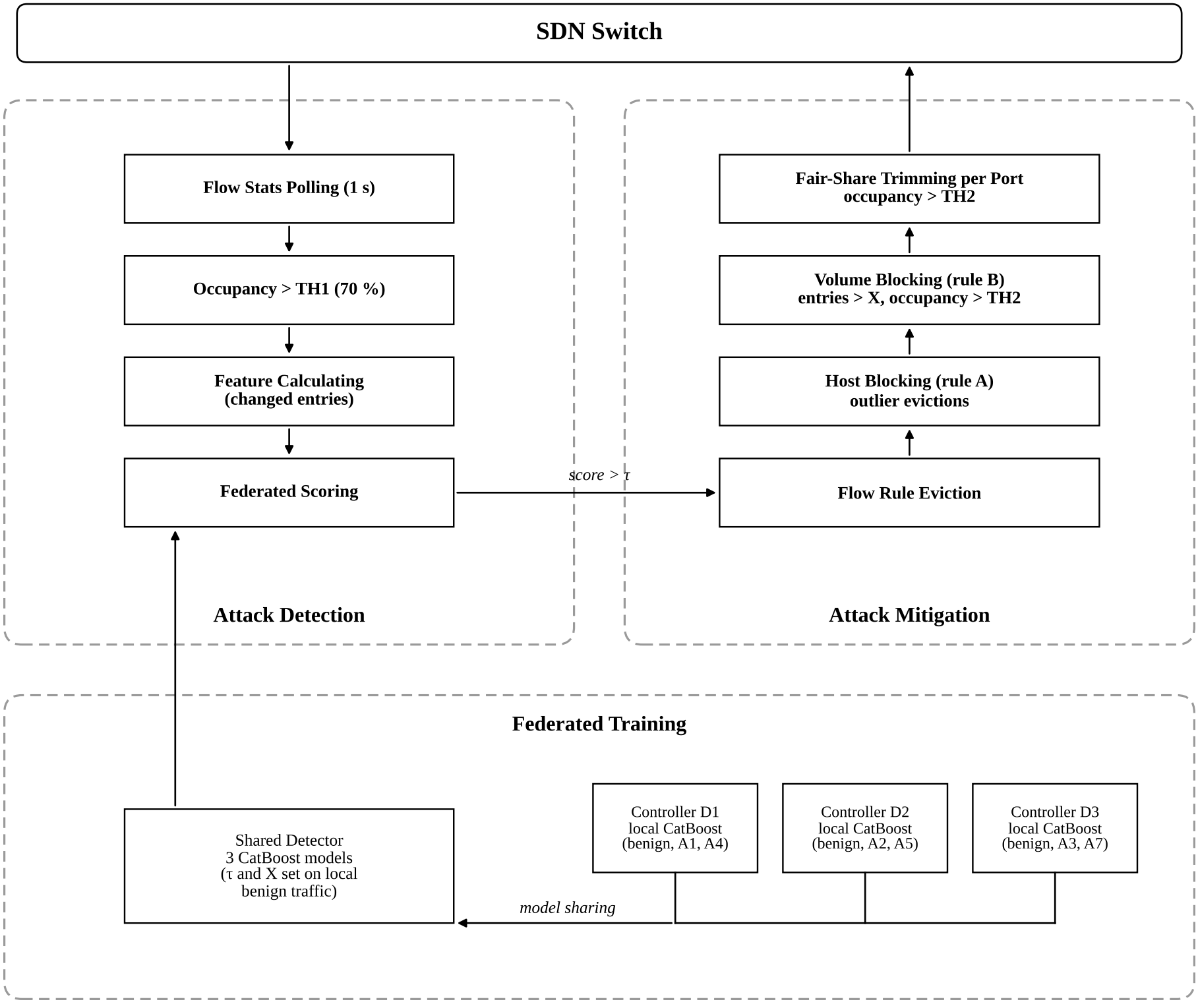
SDN Switch
Flow Stats Polling (1 s)
Occupancy > TH1 (70 %)
Feature Calculating
(changed entries)
Federated Scoring
score > τ
Attack Detection
Fair-Share Trimming per Port
occupancy > TH2
Volume Blocking (rule B)
entries > X, occupancy > TH2
Host Blocking (rule A)
outlier evictions
Flow Rule Eviction
Attack Mitigation
Federated Training
Shared Detector
3 CatBoost models
(τ and X set on local
benign traffic)
Controller D1
local CatBoost
(benign, A1, A4)
Controller D2
local CatBoost
(benign, A2, A5)
Controller D3
local CatBoost
(benign, A3, A7)
model sharing